Energy and Momentum Problems for Practice

Answer key

Conventions used throughout

• **Symbols (from the lessons):** $\vec{p} = m\vec{v}$ is momentum (kg·m/s). A prime (′) means *after* the collision; no prime means before. Subscripts name the object.

• **Method (from the lesson):** draw a before and after diagram, choose the + direction, then write $\vec{p}_{tot} = \vec{p}_{tot}'$, that is $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$.

• **Signs:** in one dimension every velocity carries its sign in the working ($v = -3.0$ m/s). The direction is written in square brackets only in the final answer.

• **Impulse:** $F\Delta t = \Delta\vec{p} = m\vec{v}_2 - m\vec{v}_1$.

• **Two dimensions:** east is +x and north is +y. The x-momentum and the y-momentum are conserved separately. [S 30° E] means start at south and turn 30° toward east.

• **Energy:** $E_k = \frac{1}{2}mv^2$ and $E_g = mgh$, with $E_{TOTi} = E_{TOTf}$ when there is no friction.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• $g = 9.81$ m/s^2. Every step keeps full precision; only final answers are rounded.

QUESTION 1

Spiked volleyball

1. A volleyball is spiked so that its incoming velocity of 4.0 m/s [E] is changed to an outgoing velocity of 21.0 m/s [W]. If the mass of the volleyball is 350 g and the impact time lasts for 0.10 s, determine the net force applied by the hand on the ball.

GIVEN

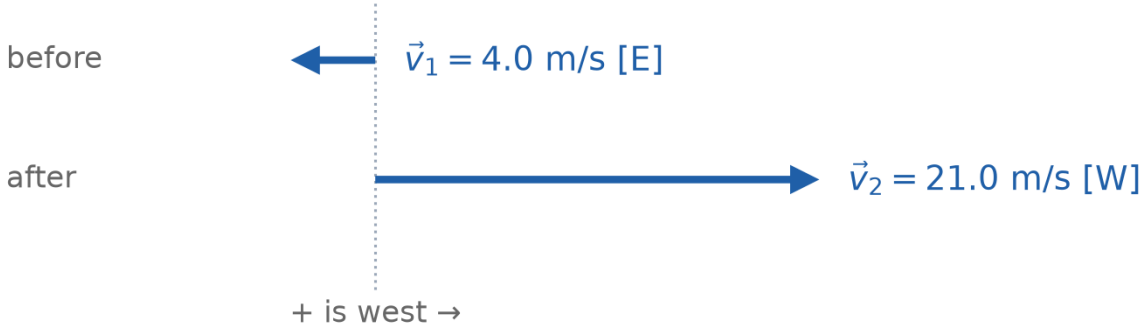
$m = 350\ \text{g} = 0.350\ \text{kg}$ $\quad v_1 = -4.0\ \text{m/s}$

$v_2 = +21.0\ \text{m/s}$ $\quad \Delta t = 0.10\ \text{s}$

$F = ?$

Sign convention: west is + (the direction of the outgoing velocity), so 4.0 m/s [E] is −4.0 m/s.

The ball's velocity before and after the spike



Impulse equals the change in momentum. Isolate F in letters first.

Formula: $F\,\Delta t = \Delta p = mv_2 - mv_1$

Divide both sides by Δt: $F = \dfrac{m(v_2 - v_1)}{\Delta t}$

Put in the numbers:

$$F = \frac{(0.350)(21.0 - (-4.0))}{0.10} = \frac{(0.350)(25.0)}{0.10} = +87.5 \text{ N}$$

F is positive, so the force points west.

FINAL ANSWER

F = +88 N

$\vec{F}$ = 88 N [W]

Worksheet answer: 87.5 N [W]
Why the answers differ: Same value, different rounding. (0.350)(25.0)/0.10 = 87.5 N. The incoming speed (4.0 m/s) and the time (0.10 s) have 2 significant figures, so the key gives 88 N; the worksheet keeps 3.

QUESTION 2

Golf ball off a wall

2. A golf ball travelling at 45 m/s [S30^{0}E] strikes a wall and rebounds such that its velocity becomes 45 m/s [N30^{0}E]. If the golf ball has a mass of 45 g, determine the impulse that has been delivered to it.

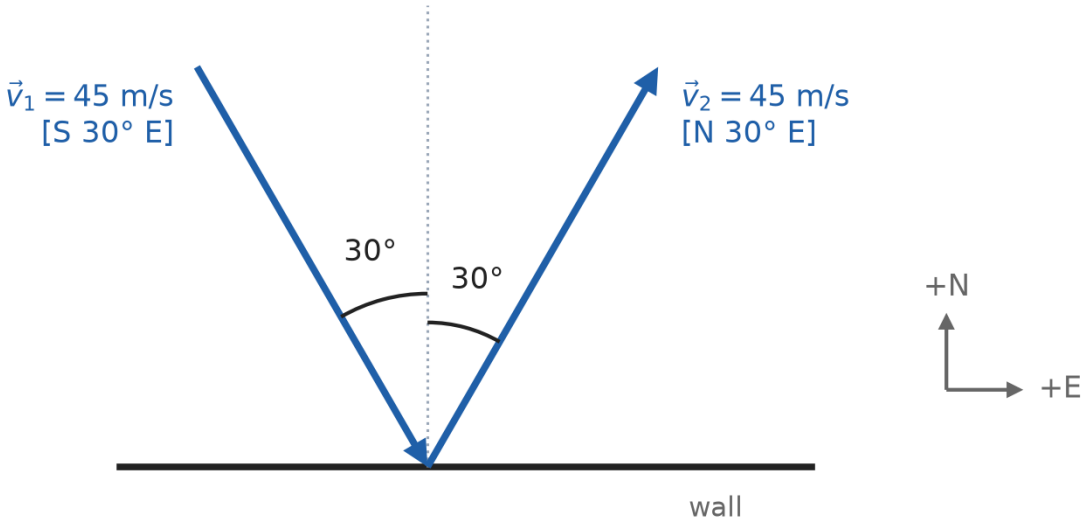
GIVEN

$m = 45\ \text{g} = 0.045\ \text{kg}$ $\qquad$ $\vec{v}_1 = 45\ \text{m/s}\ [\text{S}\,30°\,\text{E}]$

$\vec{v}_2 = 45\ \text{m/s}\ [\text{N}\,30°\,\text{E}]$ $\qquad$ $\vec{J} = ?$

Sign convention: east is +x and north is +y.

The ball's velocity before and after it hits the wall



Step 1 – Components of each velocity

Both angles are measured from the north–south line, so the east component uses sine and the north–south component uses cosine.

Before, x: $v_{1x} = +45\sin 30° = +22.500\ \text{m/s}$

Before, y: $v_{1y} = -45\cos 30° = -38.971\ \text{m/s}$

After, x: $v_{2x} = +45\sin 30° = +22.500\ \text{m/s}$

After, y: $v_{2y} = +45\cos 30° = +38.971\ \text{m/s}$

Step 2 – Impulse in each direction

Impulse equals the change in momentum, component by component.

Formula: $\vec{J} = \Delta\vec{p} = m\vec{v}_2 - m\vec{v}_1$

x: $J_x = m(v_{2x} - v_{1x}) = (0.045)(22.500 - 22.500) = 0$

y: $J_y = m(v_{2y} - v_{1y}) = (0.045)(38.971 - (-38.971))$

$J_y = +3.5074$ kg·m/s

The east component of the velocity does not change, so the impulse is entirely in +y: north.

FINAL ANSWER

$\vec{J}$ = 3.5 kg·m/s [N]

QUESTION 3

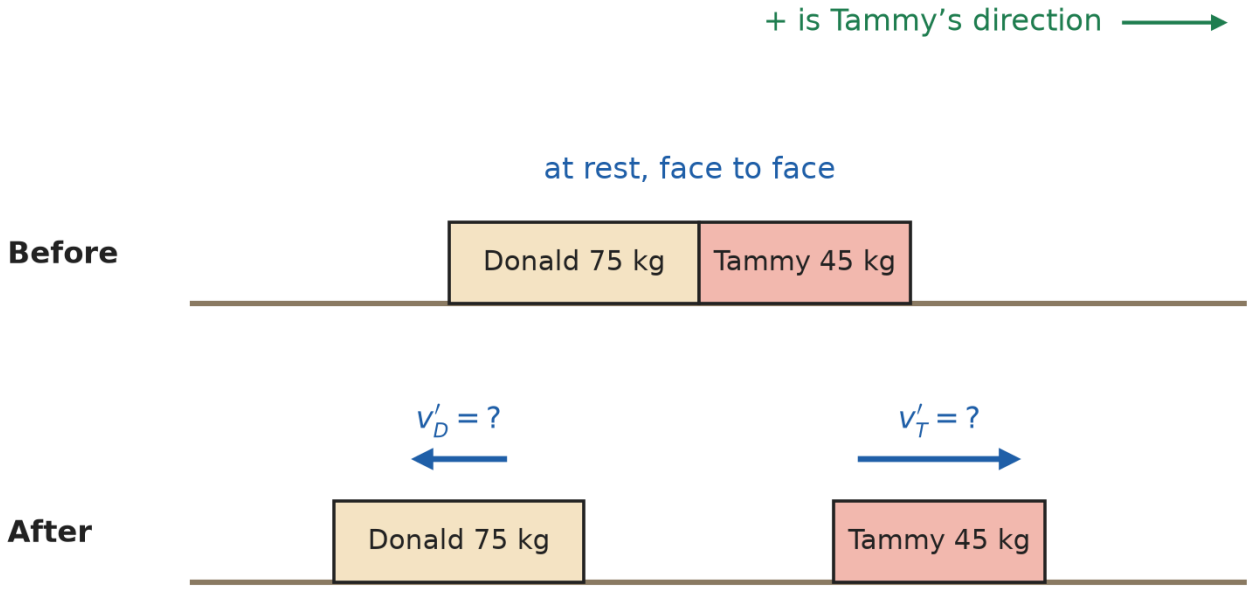
Donald and Tammy push off

3. Donald and Tammy are wearing harnesses and are hanging from the ceiling by means of ropes attached to them. They are face to face (woohoo!) and push off against each other. Donald has a mass of 75 kg and Tammy has a mass of 45 kg. Following the push, Donald swings up to a height of 0.80 m above his starting point. To what height above her starting point does Tammy rise?

GIVEN

$m_D = 75$ kg $\quad m_T = 45$ kg $\quad h_D = 0.80$ m $\quad h_T = ?$

The push (+ is the direction Tammy moves)



There are three stages. During the push momentum is conserved (the push is over quickly, and the only important horizontal forces are the ones they put on each other). During each swing afterward energy is conserved: the rope's tension is perpendicular to the motion, so it does no work.

Step 1 – Donald's speed right after the push (energy)

All of his kinetic energy at the bottom becomes gravitational energy at the top of his swing. His mass cancels.

Formula: $\frac{1}{2}m_D v'^{\,2}_D = m_D g h_D$

Isolate v'_D: $v'_D = \sqrt{2gh_D} = \sqrt{2(9.81)(0.80)} = 3.9618$ m/s

Step 2 – Tammy's speed right after the push (momentum)

They start at rest, so the total momentum is zero. With + in Tammy's direction, Donald's velocity is −3.9618 m/s.

Formula: $0 = m_D v'_D + m_T v'_T$

Isolate v'_T: $v'_T = -\frac{m_D v'_D}{m_T} = -\frac{(75)(-3.9618)}{45} = +6.6030$ m/s

Step 3 – Tammy's height (energy)

Formula: $\frac{1}{2}m_T v_T'^{\,2} = m_T g h_T$

Isolate h_T: $h_T = \frac{v_T'^{\,2}}{2g} = \frac{(6.6030)^2}{2(9.81)} = 2.2222 \text{ m}$

Tammy is lighter, so she leaves the push faster and rises higher: $h_T = (m_D/m_T)^2 h_D$.

FINAL ANSWER

h_T = 2.2 m

QUESTION 4

Fireworks rocket

4. A fireworks rocket is moving at a velocity of 50 m/s [N]. The rocket suddenly breaks into two pieces of equal mass that fly off with velocities v_1'[N30^{0}W] and v_2' [N60^{0}E]. Determine the magnitudes of v_1' and v_2'.

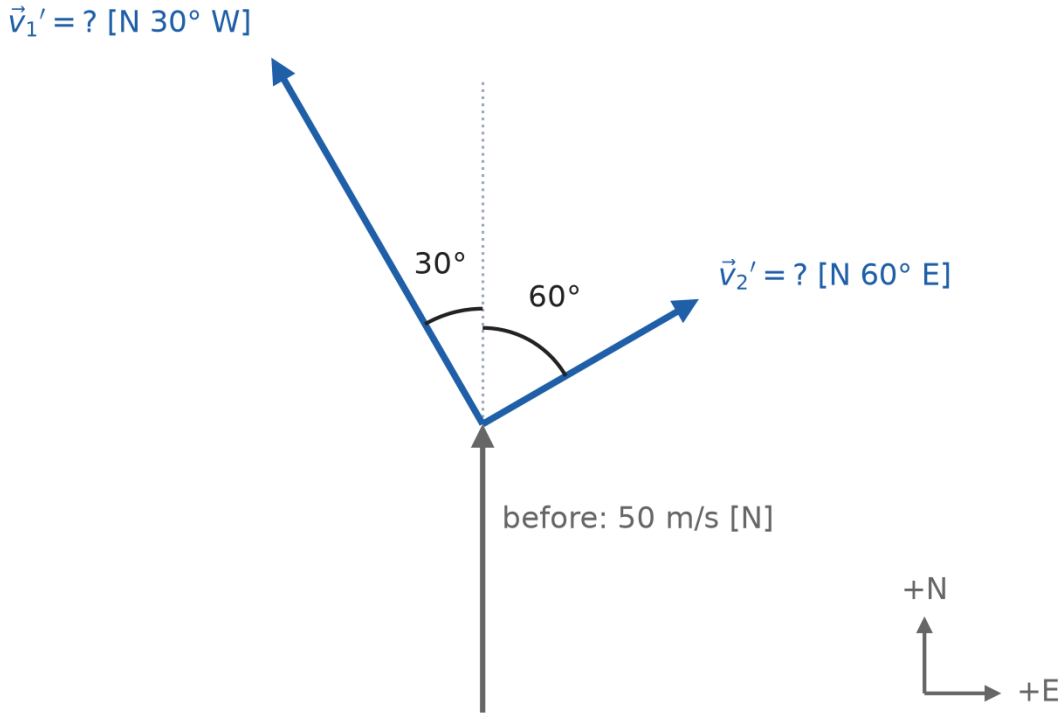
GIVEN

$\vec{v} = 50 \text{ m/s [N]}$ $\quad$ $m_1 = m_2 = m$

$\vec{v}_1{}' = ?\ [\text{N } 30° \text{ W}]$ $\quad$ $\vec{v}_2{}' = ?\ [\text{N } 60° \text{ E}]$

Sign convention: east is +x and north is +y. Each piece has mass m, so the whole rocket has mass 2m.

The rocket's velocity before, and the two pieces after



Step 1 – x-momentum (east–west)

Before the break there is no x-momentum. Piece 1 goes west (−) and piece 2 goes east (+). Both angles are measured from north, so the x-components use sine. m cancels.

Formula: $0 = -mv_1' \sin 30° + mv_2' \sin 60°$

Isolate v_1': $v_1' = \dfrac{v_2' \sin 60°}{\sin 30°} = 1.7321\, v_2'$

Step 2 – y-momentum (north–south)

Before the break the y-momentum is (2m)(50). m cancels again.

Formula: $(2m)(50) = mv_1' \cos 30° + mv_2' \cos 60°$

Divide by m: $100 = v_1' \cos 30° + v_2' \cos 60°$

Put in Step 1: $100 = (1.7321\, v_2')(0.8660) + v_2'(0.5000)$

Collect: $100 = 1.5000\, v_2' + 0.5000\, v_2' = 2.0000\, v_2'$

Isolate v_2': $v_2' = \dfrac{100}{2.0000} = 50.00$ m/s

Step 3 – Piece 1

From Step 1: $v_1' = 1.7321\, v_2' = (1.7321)(50.00) = 86.60$ m/s

FINAL ANSWER

v_1' = 87 m/s

v_2' = 50. m/s

QUESTION 5

Broken plate

5. By accident, a large plate is dropped and it breaks into three pieces. The pieces fly apart parallel to the floor. As the plate falls, it has only a vertical component and no component parallel to the floor. After the collision, the component of the total momentum parallel to the floor must remain zero since the net external force acting on the plate has no component parallel to the floor. The first piece of the plate m_1 flies off with a velocity of 3.0 m/s [N25^{0}W]. The second piece m_2 flies off with a velocity of 1.79 m/s [NE]. The third piece of mass m_3 = 1.3 kg flies off with a velocity of 3.07 m/s [S]. Determine m_1 and m_2.

GIVEN

$\vec{v}_1' = 3.0$ m/s [N 25° W]

$\vec{v}_2' = 1.79$ m/s [NE]

$m_3 = 1.3$ kg

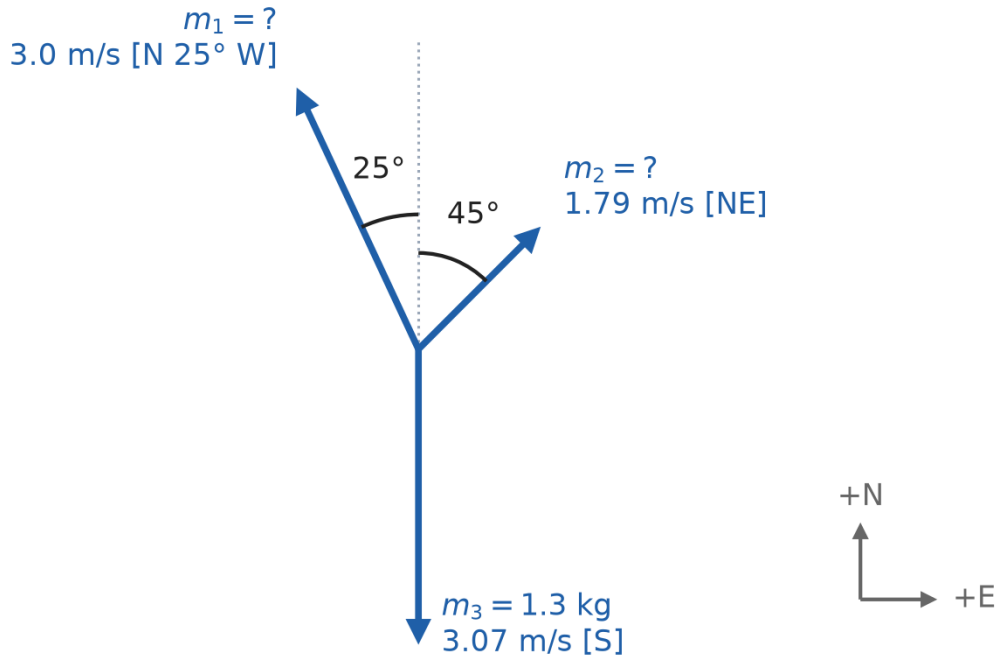
$\vec{v}_3' = 3.07$ m/s [S]

$m_1 = ?$

$m_2 = ?$

Sign convention: east is +x and north is +y, looking down at the floor. [NE] is 45° between north and east.

The three velocities, seen from above



Step 1 – x-momentum (east–west)

The total momentum parallel to the floor is zero. Piece 1 goes west (−), piece 2 goes east (+), and piece 3 has no x-component.

Formula: $0 = -m_1(3.0)\sin 25° + m_2(1.79)\sin 45°$

Put in the sines: $0 = -1.2679\,m_1 + 1.2657\,m_2$

Isolate m_2: $m_2 = \dfrac{1.2679}{1.2657}\,m_1 = 1.0017\,m_1$

Step 2 – y-momentum (north–south)

Pieces 1 and 2 go north (+) and piece 3 goes south (−).

Formula: $0 = m_1(3.0)\cos 25° + m_2(1.79)\cos 45° - (1.3)(3.07)$

Put in the cosines: $0 = 2.7189\,m_1 + 1.2657\,m_2 - 3.991$

Put in Step 1:

$3.991 = 2.7189\,m_1 + (1.2657)(1.0017)\,m_1 = 3.9868\,m_1$

Isolate m_1: $m_1 = \dfrac{3.991}{3.9868} = 1.0011$ kg

Step 3 – Piece 2

From Step 1: $m_2 = 1.0017\, m_1 = (1.0017)(1.0011) = 1.0027$ kg

FINAL ANSWER

m_1 = 1.0 kg

m_2 = 1.0 kg